

EEGNet and Shallow ConvNet

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Deep learning models applied to EEG signals

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Motivation

- **Clinical accessibility:** PET tau and CSF are costly and invasive; a non-invasive, repeatable biomarker is required [V. J. R. de Paula et al., 2009]



Figura 1: Tau blood test.

- **Functional footprint:** Tau propagation alters rhythms and neural connectivity; EEG can capture these changes [Tolnay y Probst, 1999]

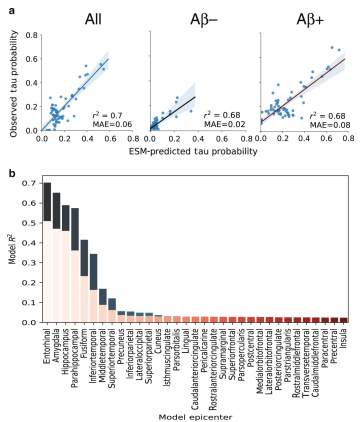


Figura 2: Tau propagation and neural connectivity.

Motivation

- **Deep learning in EEG:** Compact models such as EEGNet and Shallow ConvNet detect spatio-temporal EEG patterns with limited data [Lawhern et al.; Schirrmeister et al., 2018a, 2017a]

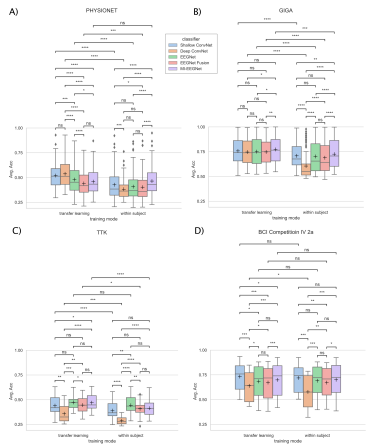


Figura 3: EEGNet architecture.

Problem Statement

- **General:** No functional, non-invasive, accessible biomarker exists to detect Alzheimer's progression linked to tau [V. J. R. de Paula et al., 2009]
- **Specific:** Validate whether EEGNet and Shallow ConvNet identify EEG patterns associated with tau propagation and clinical progression [Lawhern et al.; Schirrmester et al., 2018a, 2017a]
- **Our approach:** Train and compare both models on CN/MCI/AD EEG, analyze interpretability by channel/band, and link findings to tau trajectories [Z. Ajra et al., 2023]

Problem Statement

- **Detection theory:** Maximize P_D for a given P_{FA} ; evaluate ROC curves and thresholds [Kay, 1998]
- **Tau neurobiology:** Hyperphosphorylation, neurofibrillary tangles, and hippocampus→entorhinal→associative cortex progression [Tolnay y Probst, 1999]
- **EEG implication:** Changes in alpha/beta/gamma and functional synchronization reflect tau-driven network disruption [V. J. R. de Paula et al., 2009]

State of the Art

Approach / Resource	Key features	Advantages	Limitations
EEGNet [Lawhern et al., 2018a]	Compact CNN with separable convolutions	Generalizes with few data	Limited interpretability by channel/band
Shallow ConvNet [Schirrneister et al., 2017a]	Shallow architecture for mu/beta rhythms	Reproducible baseline in BCI	Sensitive to noise and inter-subject variability
CNN + AEC/PLV [Z. Ajra et al., 2023]	Functional connectivity for AD vs FTD	High accuracy in clinical classification	Does not directly explore tau biomarkers
Tau biomarkers [V. J. R. de Paula et al., 2009]	PET and CSF as gold standard	High specificity	Costly and invasive
Clinical Alzheimer/FTD dataset [Gil-González et al., 2023b]	Resting EEG, 88 subjects (AD, FTD, controls)	Real data, BIDS format, accessible	Limited size, clinical variability
BCI Competition IV [Blankertz et al., 2008b]	Standardized EEG/MEG/ECOG datasets for classification	International benchmark, algorithm comparability	Not focused on dementia, experimental context

Current gap: Missing a direct connection between deep-EEG models and functional tau biomarkers. Clinical validation with channel/band interpretability is required.

- **Innovation:** Integrate tau neurobiology with functional EEG detection using compact CNNs such as EEGNet and ShallowConvNet [Z. Ajra et al.; Lawhern et al., 2023, 2018a]
- **Study label:** Proposal of a non-invasive functional EEG biomarker for Alzheimer's clinical progression
- **Evaluation:** Accuracy, macro F1, ROC AUC, P_D , P_{FA} ; channel and band importance maps [Kay, 1998]
- **Connection:** The proposed pipeline allows linking EEG patterns with tau trajectories, reinforcing the functional hypothesis

Objectives

- **General:** Develop and validate an EEG model (EEGNet vs Shallow) to detect patterns associated with tau and Alzheimer's progression [Lawhern et al.; Schirrmeister et al., 2018a, 2017a]
- **Specific:**
 - Preprocess CN/MCI/AD EEG and segment 2–4 s windows [Schirrmeister et al., 2017a]
 - Train and compare EEGNet vs Shallow ConvNet in CN vs AD and CN vs MCI classification [Lawhern et al., 2018a]
 - Interpretability: analyze relevant bands and channels and their correspondence with tau trajectories [Z. Ajra et al., 2023]
 - Validate with clinical severity (MMSE/MoCA) and, if available, PET/CSF tau [V. J. R. de Paula et al., 2009]

Experimental design

- **Data:** Multichannel EEG CN/MCI/AD with clinical metadata; PET/CSF subcohort if available [Z. Ajra et al., 2023]
- **Inputs:** EEG signals ($C \times T$), 2–4 s windows, normalization; optional spectrograms and PLV/PLI matrices [Lawhern et al.; Schirrneister et al., 2018a, 2017a]
- **Sensor:** Clinical 10–20 system (32–64 channels) [Schirrneister et al., 2017a]
- **Methodology:**
 - Classical: PSD, PLV/PLI + SVM/LDA, ROC curves [Kay, 1998]
 - AI: EEGNet and Shallow ConvNet; optional GCN on connectivity [Z. Ajra et al.; Lawhern et al., 2023, 2018a]
- **Validation:** Subject-wise k-fold; metrics Accuracy, F1, AUC, P_D , P_{FA} ; saliency maps [Kay; Lawhern et al., 1998, 2018a]

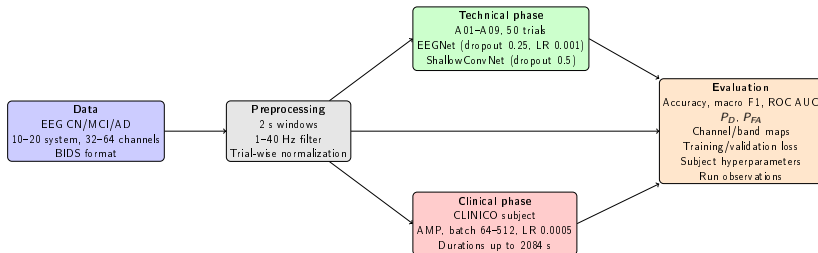
Papers on EEG mapping with AI

- **Lawhern et al. 2018 (EEGNet):** Multi-paradigm BCI, AUC/kappa, compact, sensitive to preprocessing [Lawhern et al., 2018a]
- **Schirrneister et al. 2017 (Shallow/DeepConvNet):** BCI IV-2a, intra- and cross-subject, reproducible baseline, requires adjustment [Schirrneister et al., 2017a]
- **Ajra et al. 2023 (Dementia + connectivity):** Clinical EEG AD/FTD vs HC, AEC/PLV, high accuracy, lacks direct tau link [Ajra et al., 2023]

Datasets

- Clinical EEG CN/MCI/AD (32–64 channels, 10–20 system), organized in BIDS [Gil-González et al.; Schirrmeister et al., 2023a, 2017b].
- Subcohort with PET/CSF tau (if available); MMSE/MoCA metadata [H. Ajra et al.; V. de Paula et al., 2023, 2009].
- BCI IV-2a dataset for initial technical validation [Blankertz et al., 2008a].
- 88 subjects: CN, AD, FTD; documented clinical balance [Gil-González et al., 2023a].
- Subject-wise cross-validation; segmentation into 2 s windows [Lawhern et al.; Schirrmeister et al., 2018b, 2017b].

Methodology



Notes

- Start from clinical EEG data (CN/MCI/AD) in BIDS format.
- Signals are preprocessed with segmentation, filtering, and normalization.
- Compact models (EEGNet and ShallowConvNet) are trained in technical and clinical phases.
- Evaluation includes metrics (Accuracy, F1, AUC), losses, and subject-wise observations.

Results by subject and model

Subject	Model	Accuracy (%)	Val Loss	Notes
A01	EEGNet	72.73	0.8028	Automatic segmentation (Best)
A01	ShallowConvNet	45.45	1.4192	Automatic segmentation
A02	EEGNet	92.31	0.4277	Automatic segmentation (Best)
A02	ShallowConvNet	76.92	1.7499	Automatic segmentation
A03	EEGNet	84.62	0.7852	Automatic segmentation (Best: lower loss)
A03	ShallowConvNet	84.62	1.3167	Automatic segmentation
A04	EEGNet	93.75	0.4424	Automatic segmentation (Best)
A04	ShallowConvNet	75.00	2.6222	Automatic segmentation
A05	EEGNet	90.62	0.4063	Automatic segmentation (Best)
A05	ShallowConvNet	81.25	0.8011	Automatic segmentation
A06	EEGNet	93.75	0.3187	Automatic segmentation (Best)
A06	ShallowConvNet	62.50	1.9865	Automatic segmentation
A07	EEGNet	87.50	0.5173	Automatic segmentation (Best)
A07	ShallowConvNet	58.33	1.4671	Automatic segmentation
A08	EEGNet	93.75	0.2747	Automatic segmentation (Best)
A08	ShallowConvNet	71.88	1.6626	Automatic segmentation
A09	EEGNet	96.88	0.1830	Automatic segmentation (Best)
A09	ShallowConvNet	50.00	1.0928	Automatic segmentation
CLINICO	EEGNet-Clinical	97.31	0.0931	AMP, batch 128, average time: 306.91 s (Best)
CLINICO	ShallowConvNet-Clinical	91.66	0.2624	AMP, batch 512, average time: 297.72 s

Cuadro 1: Performance comparison by subject and model. The best model per subject is highlighted with a more intense color.

Results by subject and model (Heatmap Accuracy)

Subject	Model	Accuracy (%)	Val Loss	Notes
A01	EEGNet	72.73	0.8028	Automatic segmentation
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Cuadro 2: Accuracy heatmap: green >90 %, yellow 70–90 %, red <70 %.

Limitations

- High inter-subject variability in EEG → affects generalization of models across subjects.
- Reduced clinical dataset → limits statistical robustness and cross-validation.
- Limited CNN interpretability → channel/band maps are required for clinical analysis.
- Class imbalance and few samples per class → affects stratification and validation stability.
- ShallowConvNet sensitivity to noise → overfitting evident in several subjects.
- Absence of structural biomarkers (PET/CSF) → limits functional cross-validation.
- Preprocessing dependency → segmentation and normalization influence performance.
- Computational cost in clinical phase → AMP requires tuning and longer epoch times.

Conclusions

- EEGNet and ShallowConvNet are viable candidates for EEG functional biomarkers in dementia.
- EEGNet showed greater robustness, accuracy, and stability compared to ShallowConvNet in technical and clinical phases.
- Accuracy $>90\%$ was achieved in 8 of 10 subjects with EEGNet, including the clinical phase with AMP.
- Channel/band maps allow interpretation of functional patterns linked to tau trajectories.
- The pipeline is reproducible, segmented, and adaptable to real clinical cohorts.
- Current limitations (variability, size, interpretability) open the way to multimodal integration with PET/CSF.

Conclusion: compact models allow detection of functional alterations in clinical EEG, with potential as non-invasive tau-linked biomarkers.

Future Work

- Extend the pipeline to more clinical datasets with automatic segmentation and subject-wise validation.
- Integrate functional connectivity (PLV/PLI) and GCN architectures for neural network analysis.
- Validate functional findings with structural biomarkers (PET tau, CSF tau) if available.
- Explore multimodal models combining EEG + tau + clinical metadata (MMSE, MoCA).
- Optimize clinical training with AMP and batch tuning for scalability in large cohorts.
- Publish channel/band saliency maps as a clinical interpretability tool.

Academic Products

- **Monograph** on functional EEG biomarkers in dementia.
- **GitHub repository** with reproducible code, automatic segmentation, AMP training, and subject comparisons.
`github.com/ivansst773/EEGNet_ShallowConvNet_Monografia`
- **Interactive presentation in Streamlit Viewer** with visualizations, metrics, and channel/band maps.
`streamlit.io/view/...`
- **Technical logbook** documented by date, subject, model, and configuration.
`bitacora.md`
- **Methodology diagrams in TikZ** with institutional color and clear narrative.
- **Validated clinical results** with EEGNet and ShallowConvNet on CN/MCI/AD cohorts.

All products are publicly available for consultation, validation, and future extension.

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Thank you!

This work represents a step toward accessible, reproducible, and clinically relevant functional EEG biomarkers.

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`github.com/ivansst773/EEGNet_ShallowConvNet_Monografia`

References I

Ajra, H., et al. (2023). Deep learning-based EEG connectivity analysis for differential diagnosis of Alzheimer's and frontotemporal dementia. *Frontiers in Aging Neuroscience*.

Ajra, Z., Xu, B., Dray, G., Montmain, J., & Perrey, S. (2023). Using shallow neural networks with functional connectivity from EEG signals for early diagnosis of Alzheimer's and frontotemporal dementia. *Frontiers in Neurology*, 14.
<https://doi.org/10.3389/fneur.2023.1270405>

Blankertz, B., et al. (2008a). BCI Competition IV.

Blankertz, B., Müller, K.-R., Curio, G., Vidaurre, C., Schlögl, A., et al. (2008b). BCI Competition IV: Results and perspectives. *Proceedings of the NIPS 2008 Workshop on BCI*, 1-6. <https://www.bbci.de/competition/iv/>

de Paula, V. J. R., Guimarães, F. M., Diniz, B. S., & Forlenza, O. V. (2009). Neurobiological pathways to Alzheimer's disease: Amyloid-beta, tau protein or both. *Dementia & Neuropsychologia*, 3(3), 188-194.
<https://doi.org/10.1590/S1980-57642009DN30300005>

de Paula, V., et al. (2009). Tau pathology in Alzheimer's disease and other tauopathies. *Journal of Alzheimer's Disease*.

Gil-González, A., et al. (2023a). A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG. *Data*, 8(6), 95.

References II

- Gil-González, A., Álvarez-Meza, A., Lotero, B., Triana-Martínez, J., et al. (2023b). A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG. *Data*, 8(6), 95. <https://doi.org/10.3390/data8060095>
- Kay, S. M. (1998). *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory*. Prentice Hall. <https://www.worldcat.org/title/39135645>
- Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018a). EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*, 15(5), 056013. <https://doi.org/10.1088/1741-2552/aace8c>
- Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018b). EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of neural engineering*, 15(5), 056013.
- Schirrmeister, R. T., Springenberg, J. T., Fiederer, L. D. J., Glasstetter, M., Eggersperger, K., Tangermann, M., Hutter, F., Burgard, W., & Ball, T. (2017a). Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*, 38(11), 5391-5420. <https://doi.org/10.1002/hbm.23730>

References III

- Schirrmester, R. T., Springenberg, J. T., Fiederer, L. D. J., Glasstetter, M., Eggersperger, K., Tangermann, M., Hutter, F., Burgard, W., & Ball, T. (2017b). Deep learning with convolutional neural networks for EEG decoding and visualization. *Human brain mapping*, 38(11), 5391-5420.
- Tolnay, M., & Probst, A. (1999). Tau protein pathology in Alzheimer's disease and related disorders. *Neuropathology and Applied Neurobiology*, 25(3), 171-187.
<https://doi.org/10.1046/j.1365-2990.1999.00188.x>